

GSM8K 30-item Spot-Check

사용자(2차 annotator) 검토 가이드

이 PDF는 무엇인가?

논문 §4의 GSM8K "125-item equivalence-verified subset" 주장을 강화하기 위해, 30개 item을 두 명의 annotator(=평가자)가 독립적으로 평가합니다. 1차 평가자(Claude AI)는 이미 30개 모두 평가 완료. 사용자께서 그 중 5개의 "애매한" item만 직접 평가하시면 됩니다.

왜 5개만?

나머지 25개는 1차 평가자가 "명백히 동등(equivalent)"으로 판정해서 사용자 검토가 거의 의미 없습니다. 5개는 reviewer(논문 심사자)가 공격할 가능성이 있는 borderline item들입니다. 이 5개에 대한 사용자 의견이 §4 본문 한 문장으로 들어가서 reviewer 방어가 강화됩니다.

각 item에 대해 무엇을 결정하는가?

orig 문제 ↔ pert 문제 두 개를 비교해서, 두 문제가 "같은 어려움/같은 풀이 구조"인지 한 단어로 답합니다. 세 가지 단어 중 하나:

라벨	의미	예시 판단 기준
equivalent (동등)	두 문제가 거의 같은 어려움 + 같은 풀이 구조	예: 완전 종속의 연산(+/-/×/÷) 같은 단계 수 비슷한 자릿수/난이도
partial (부분 동등)	구조는 같지만 한쪽이 미세하게 다름	예: 한쪽은 정수, 다른 쪽은 소수 등장 예: 한쪽은 carry 많고 다른 쪽은 적음
non-equivalent (비동등)	구조나 어려움이 분명히 다름	예: 단계 수가 다름 예: 한쪽이 훨씬 더 어려움

5개 item을 다 보신 후 답변 양식

한 줄로: "43=equivalent, 47=equivalent, 56=partial, 57=equivalent, 102=equivalent"

이 결과를 알려주시면 §4에 한 문장 추가하고 PDF 재컴파일합니다.

1차 평가자(Claude) 의견 요약

idx 43 = equivalent idx 47 = equivalent idx 56 = partial (소수점 1.5/40.5 등장) idx 57 = equivalent
idx 102 = equivalent

→ 사용자 의견이 1차 평가자와 같으면 그대로, 다르면 사용자 의견 우선.

Item 1 of 5 · idx 43

왜 검토가 필요한가: 8단계 — 모든 step이 정말 1:1 parallel한가?

ORIG (원본 문제)

문제: Mike decides he wants to replace his movie collection with digital versions. He has 600 movies. A third of the movies are in various series and he knows he can get those for only \$6 of the cost of a normal movie by just buying the series together. 40% of the remaining movies are older movies which are \$5. How much does replacing the movies cost if a normal movie costs \$10?

풀이/정답:

There are $600/3 = \langle\langle 600/3 = 200 \rangle\rangle 200$ movies in series

So he pays $200 * 6 = \$\langle\langle 200 * 6 = 1200 \rangle\rangle 1200$

He has $600 - 200 = \langle\langle 600 - 200 = 400 \rangle\rangle 400$ movies not in series

$400 * .4 = \langle\langle 400 * .4 = 160 \rangle\rangle 160$ movies are old

So those cost $160 * 5 = \$\langle\langle 160 * 5 = 800 \rangle\rangle 800$

$400 - 160 = \langle\langle 400 - 160 = 240 \rangle\rangle 240$ movies are neither old nor in series

They cost $240 * 10 = \$\langle\langle 240 * 10 = 2400 \rangle\rangle 2400$

So the total cost is $1200 + 800 + 2400 = \$\langle\langle 1200 + 800 + 2400 = 4400 \rangle\rangle 4400$

4400

PERT (변형 문제 — 숫자만 바뀜)

문제: Sarah decides she wants to replace her book collection with digital versions. She has 480 books. A quarter of the books are in various series and she knows she can get those for only \$8 of the cost of a normal book by just buying the series together. 50% of the remaining books are classic novels which are \$7. How much does replacing the books cost if a normal book costs \$12?

풀이/정답:

There are $480/4 = \langle\langle 480/4 = 120 \rangle\rangle 120$ books in series. So she pays $120 * 8 = \$\langle\langle 120 * 8 = 960 \rangle\rangle 960$. She has

$480 - 120 = \langle\langle 480 - 120 = 360 \rangle\rangle 360$ books not in series. $360 * .5 = \langle\langle 360 * .5 = 180 \rangle\rangle 180$ books are classic

novels. So those cost $180 * 7 = \$\langle\langle 180 * 7 = 1260 \rangle\rangle 1260$. $360 - 180 = \langle\langle 360 - 180 = 180 \rangle\rangle 180$ books are

neither classic nor in series. They cost $180 * 12 = \$\langle\langle 180 * 12 = 2160 \rangle\rangle 2160$. So the total cost is

$960 + 1260 + 2160 = \$\langle\langle 960 + 1260 + 2160 = 4380 \rangle\rangle 4380$.

4380

여기서 사용자 결정:

위 두 문제(ORIG vs PERT)를 비교했을 때:

- 같은 종류의 연산을 같은 순서로 사용하는가?
- 비슷한 자릿수/소수/받아올림 등 비슷한 어려움인가?

→ 한 단어로 답: equivalent / partial / non-equivalent

Item 2 of 5 · idx 47

왜 검토가 필요한가: abstract 첫 문장 anecdote의 원본 — 정말 elementary arithmetic으로 풀리는지

ORIG (원본 문제)

문제: Jerry is twice as old as he was 5 years ago. How old will Jerry be in 3 years?

풀이/정답:

We know that Jerry's Age/2 = Jerry's Age - 5

Therefore Jerry's Age = 2 Jerry's age - 10

Therefore -Jerry's Age = - 10

Therefore Jerry's Age = $\langle\langle 10=10 \rangle\rangle 10$

In three years Jerry will be 13 because $10 + 3 = \langle\langle 10+3=13 \rangle\rangle 13$

13

PERT (변형 문제 — 숫자만 바뀜)

문제: Megan is three times as old as she was 4 years ago. How old will Megan be in 5 years?

풀이/정답:

We know that Megan's Age/3 = Megan's Age - 4. Therefore, Megan's Age = 3 Megan's Age - 12.

Therefore, -Megan's Age = -12. Therefore, Megan's Age = $\langle\langle 12=12 \rangle\rangle 12$. In five years, Megan will be 17 because $12 + 5 = \langle\langle 12+5=17 \rangle\rangle 17$.

17

여기서 사용자 결정:

위 두 문제(ORIG vs PERT)를 비교했을 때:

- 같은 종류의 연산을 같은 순서로 사용하는가?
- 비슷한 자릿수/소수/받아올림 등 비슷한 어려움인가?

→ 한 단어로 답: equivalent / partial / non-equivalent

Item 3 of 5 · idx 56

왜 검토가 필요한가: 1차에서 partial 판정 — 소수점 등장

ORIG (원본 문제)

문제: Bob needs to buy potting soil to fill the raised beds in his vegetable garden. He has 10 raised beds, and each bed is 2 feet wide by 8 feet long by 2 feet tall. Each bag of potting soil holds 2 cubic feet of soil and costs \$12. How much will the potting soil cost him?

풀이/정답:

Each bed is 2 feet wide by 8 feet long by 2 feet high, so they each hold $2 * 8 * 2 = \langle\langle 2*8*2=32 \rangle\rangle 32$ cubic feet of dirt.

He has 10 beds, so he will need $10 * 32 = \langle\langle 10*32=320 \rangle\rangle 320$ cubic feet of dirt.

Each bag of soil is 0.75 cubic feet, so Bob will need $320 / 2 = \langle\langle 320/2=160 \rangle\rangle 160$ bags of soil.

Each bag is \$12, so all the soil will cost him $\$12 * 160 = \$\langle\langle 12*160=1920 \rangle\rangle 1920$.

1920

PERT (변형 문제 — 숫자만 바뀜)

문제: Alice needs to buy gravel to fill the flower beds in her garden. She has 12 flower beds, and each bed is 3 feet wide by 9 feet long by 1.5 feet tall. Each bag of gravel holds 3 cubic feet of gravel and costs \$15. How much will the gravel cost her?

풀이/정답:

Each bed is 3 feet wide by 9 feet long by 1.5 feet high, so they each hold $3 * 9 * 1.5 =$

$\langle\langle 3*9*1.5=40.5 \rangle\rangle 40.5$ cubic feet of gravel. She has 12 beds, so she will need $12 * 40.5 =$

$\langle\langle 12*40.5=486 \rangle\rangle 486$ cubic feet of gravel. Each bag of gravel is 3 cubic feet, so Alice will need $486 / 3 =$

$\langle\langle 486/3=162 \rangle\rangle 162$ bags of gravel. Each bag is \$15, so all the gravel will cost her $\$15 * 162 =$

$\$\langle\langle 15*162=2430 \rangle\rangle 2430$.

2430

여기서 사용자 결정:

위 두 문제(ORIG vs PERT)를 비교했을 때:

- 같은 종류의 연산을 같은 순서로 사용하는가?
- 비슷한 자릿수/소수/받아올림 등 비슷한 어려움인가?

→ 한 단어로 답: equivalent / partial / non-equivalent

Item 4 of 5 · idx 57

왜 검토가 필요한가: orig은 $70+70$ (덧셈), pert는 50×3 (곱셈) — 같은 op로 봐야 하나?

ORIG (원본 문제)

문제: Bill is trying to figure out how many water bottles he needs to buy for his party. He started out with 2 cases of water which have 24 bottles each. He has 70 guests that will be arriving, and he wants to have 2 bottles of water for each guest. How many additional bottles of water will he need to buy?

풀이/정답:

Bill has 2 cases of 24 bottles each, so he has $24 * 2 = \langle\langle 2 * 24 = 48 \rangle\rangle 48$ bottles of water

Bill needs enough water for 70 guests to have 2 bottles each, so he needs $70 * 2 = \langle\langle 70 * 2 = 140 \rangle\rangle 140$ bottles of water.

Bill needs 140 bottles of water and has 48, so he needs to buy $140 - 48 = \langle\langle 140 - 48 = 92 \rangle\rangle 92$ bottles of water.

92

PERT (변형 문제 — 숫자만 바뀜)

문제: Samantha is trying to figure out how many juice boxes she needs to buy for her picnic. She started out with 3 cases of juice which have 20 boxes each. She has 50 friends that will be arriving, and she wants to have 3 juice boxes for each friend. How many additional juice boxes will she need to buy?

풀이/정답:

Samantha has 3 cases of 20 boxes each, so she has $20 * 3 = \langle\langle 3 * 20 = 60 \rangle\rangle 60$ juice boxes. Samantha needs enough juice for 50 friends to have 3 boxes each, so she needs $50 * 3 = \langle\langle 50 * 3 = 150 \rangle\rangle 150$ juice boxes. Samantha needs 150 juice boxes and has 60, so she needs to buy $150 - 60 = \langle\langle 150 - 60 = 90 \rangle\rangle 90$ juice boxes.

90

여기서 사용자 결정:

위 두 문제(ORIG vs PERT)를 비교했을 때:

- 같은 종류의 연산을 같은 순서로 사용하는가?
- 비슷한 자릿수/소수/받아올림 등 비슷한 어려움인가?

→ 한 단어로 답: equivalent / partial / non-equivalent

Item 5 of 5 · idx 102

왜 검토가 필요한가: 자연어 + 숫자 동시 변경 — 정말 같은 구조인가?

ORIG (원본 문제)

문제: On a particular week, a tow truck pulled ten cars for each of the first three days and then four fewer cars on each of the remaining days of the week. Calculate the total number of cars it towed that week.

풀이/정답:

On the first three days, towing ten cars a day, the tow truck pulled $3 \times 10 = \langle\langle 3 \times 10 = 30 \rangle\rangle 30$ cars.

It pulled four fewer cars each day, which is $10 - 4 = \langle\langle 10 - 4 = 6 \rangle\rangle 6$ cars on the remaining days.

If it pulled ten cars for three days and four less on the remaining days, then it pulled for $7 - 3 = 4$ days, four cars less each day.

For the four days, the car pulled $6 \times 4 = \langle\langle 6 \times 4 = 24 \rangle\rangle 24$ cars.

The total number of cars it pulled that week is $24 + 30 = \langle\langle 24 + 30 = 54 \rangle\rangle 54$ cars.

54

PERT (변형 문제 — 숫자만 바뀜)

문제: During a specific week, a delivery truck transported twelve packages for each of the first four days and then five fewer packages on each of the remaining days of the week. Calculate the total number of packages it delivered that week.

풀이/정답:

On the first four days, delivering twelve packages a day, the delivery truck transported $4 \times 12 = \langle\langle 4 \times 12 = 48 \rangle\rangle 48$ packages.

It delivered five fewer packages each day, which is $12 - 5 = \langle\langle 12 - 5 = 7 \rangle\rangle 7$ packages on the remaining days.

If it delivered twelve packages for four days and five less on the remaining days, then it delivered for $7 - 4 = 3$ days, five packages less each day. For the three days, the truck delivered $7 \times 3 = \langle\langle 7 \times 3 = 21 \rangle\rangle 21$ packages. The total number of packages it delivered that week is $21 + 48 = \langle\langle 21 + 48 = 69 \rangle\rangle 69$ packages.

69

여기서 사용자 결정:

위 두 문제(ORIG vs PERT)를 비교했을 때:

- 같은 종류의 연산을 같은 순서로 사용하는가?
- 비슷한 자릿수/소수/받아올림 등 비슷한 어려움인가?

→ 한 단어로 답: equivalent / partial / non-equivalent

최종 답변 작성 예시

5개 item을 다 보셨다면, 다음 중 한 형태로 답변해주세요:

형태 1 — 항목별 라벨

43=equivalent
47=equivalent
56=partial
57=equivalent
102=equivalent

형태 2 — 1차 평가자와 다른 것만 짚기

"56과 57에 동의 안 함. 56은 non-equivalent, 57은 partial. 나머지 3개 동의."

형태 3 — 신뢰 기반 빠른 결정

"1차 평가자 결과 그대로 신뢰. 추가 변경 없음."

이 결과를 받으면 다음을 진행합니다:

- Cohen's κ (annotator 일치도) 계산
- §4 fidelity caveat에 한 문장 추가 (예: "30 items 중 N개가 두 평가자 모두 equivalent 판정")
- PDF 재컴파일 및 최종 검증